Commodity Intelligence & Fleet Operations Report

Data-Driven Analysis for Equipment & Operations Planning

Bull and Bear Co.

Prepared: March 2026

Data Source: QuickManage API (October 2025 – February 2026)

Classification: Confidential — Internal Use Only

Executive Summary

Background

Bull and Bear Co. operates a liquid tanker fleet that executed 3,346 trips during the analysis period (October 2025 – February 2026). Prior to this analysis, the company had no systematic method to track which commodities were being hauled across the fleet. Commodity data was buried in driver documents, bills of lading, and shipping papers — inaccessible for operational planning or equipment decisions.

Methodology

An automated extraction system was developed to scan all available driver documentation — including bills of lading (BOLs), shipping papers, delivery tickets, and safety data sheets. The system used optical character recognition (OCR) and pattern matching across 20+ document formats to identify commodity names from each trip's paperwork. The system was iteratively refined through 4 rounds of extraction, progressively improving coverage with each round.

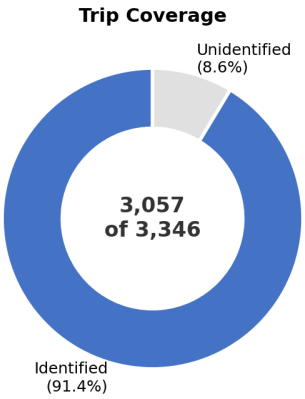
Following extraction, each identified commodity was matched to its published viscosity (cP) value using a curated database of 269 industrial chemicals, 705 trade name mappings, and PubChem as a fallback. This enables equipment-level planning based on the physical properties of the products being transported.

Key Metrics

Total Trips Analyzed	**3,346**
Trips with Commodity Identified	**3,057 (91.4%)**
Unique Commodities Discovered	**1,432**
Commodities Matched to Viscosity (cP)	**876 of 1432 (61.2%)**
Load Coverage with Viscosity Data	**2,165 of 3,057 (70.8%)**
Median Viscosity	**30 cP**

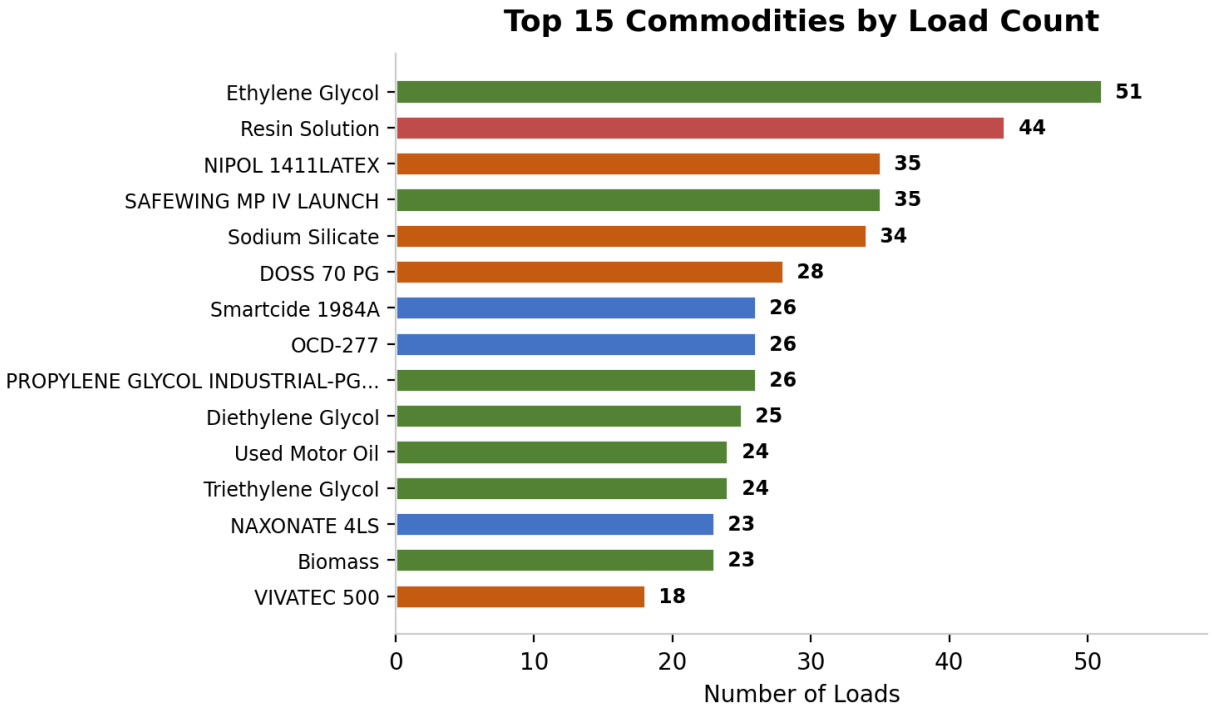
Key Findings

- Extraction coverage reached 91.4% — identifying commodities on 3,057 of 3,346 trips. This represents a major improvement from the previous 40.6% baseline.
- Median viscosity across all identified loads is 30 cP — the majority of products are low-viscosity and easy to handle.
- 90% of loads have viscosity below 200 cP, while only ~6% are high-viscosity products (500+ cP).
- The fleet hauls 1,432 distinct products, but the top 50 commodities account for the majority of loads — indicating a concentrated core portfolio.



Commodity Portfolio

The following chart shows the 15 most frequently hauled commodities during the analysis period. Bar colors indicate viscosity category: blue = water-like, green = medium, orange = heavy, red = very heavy, purple = ultra-heavy.



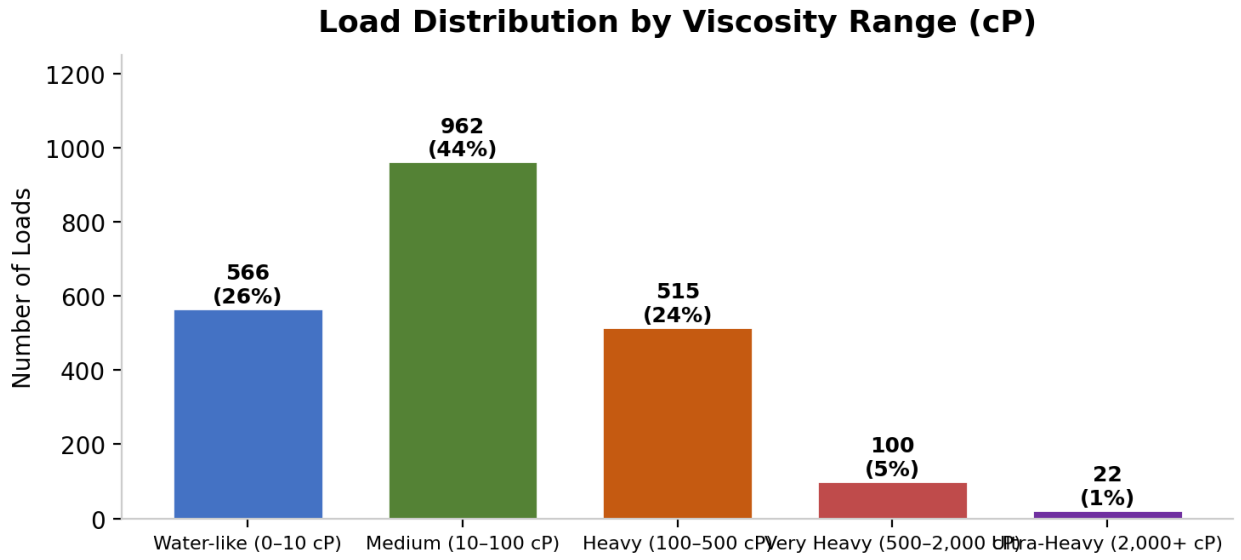
Top 15 Commodity Detail

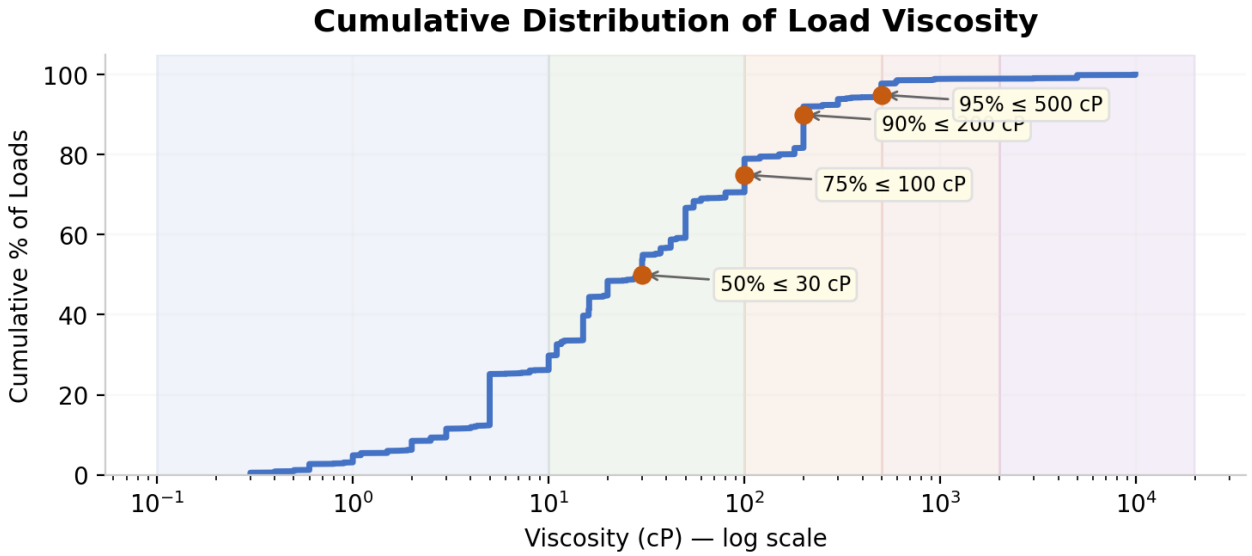
Rank	Commodity	Loads	Viscosity (cP)
1	Ethylene Glycol	51	16
2	Resin Solution	44	500
3	NIPOL 1411LATEX	35	200
4	SAFEWING MP IV LAUNCH	35	15
5	Sodium Silicate	34	180
6	DOSS 70 PG	28	200
7	Smartcide 1984A	26	5
8	OCD-277	26	1
9	PROPYLENE GLYCOL INDUSTRIAL-PGI	26	42
10	Diethylene Glycol	25	30
11	Used Motor Oil	24	20
12	Triethylene Glycol	24	37
13	NAXONATE 4LS	23	5
14	Biomass	23	50
15	VIVATEC 500	18	100

INSIGHT: The top 163 commodities (out of 1,432) represent 60% of all identified loads — a concentrated core portfolio that should drive ec

Viscosity Analysis

Viscosity (measured in centipoise, cP) describes how thick a liquid is. Lower viscosity products flow easily like water, while higher viscosity products are progressively thicker and require more specialized handling equipment.





KEY TAKEAWAY: 90% of all loads have viscosity ≤ 200 cP. 95% of loads ≤ 500 cP.

Data Coverage & Recommendations

Data Coverage Assessment

Of 3,346 trips analyzed, commodity information was successfully extracted from 3,057 trips (91.4%). This represents a significant achievement in automated document extraction. The remaining 289 trips (8.6%) could not be identified due to missing driver documents (5 trips with no files), or complex/non-standard document formats (284 trips with extraction misses).

Of the 3,057 identified trips, viscosity data was matched for 2,165 loads (70.8%). The identified subset covers 1,432 distinct commodities across all major product categories including chemicals, petroleum products, food-grade materials, solvents, surfactants, and industrial specialties. The viscosity analysis provides a solid basis for equipment and operational planning.

Recommendations

1 Mandate Commodity Entry in QuickManage

Require dispatchers to enter the commodity name for every trip at booking time. This single operational change would provide real-time visibility into what the fleet is carrying and eliminate the need for document-based extraction. Cost: zero — it is a process change only.

2 Standardize Driver Document Submission

Implement a consistent process for drivers to upload bills of lading and shipping papers within 24 hours of delivery. The 8.6% extraction gap stems primarily from missing or late documentation. Consider a mobile app integration with QuickManage for immediate upload at pickup/delivery.

3 Leverage Viscosity Data for Fleet Planning

With 70.8% of identified loads matched to viscosity values, the data now supports equipment planning. 26% of loads are water-like (under 10 cP), while 6% are high-viscosity (500+ cP). Use this distribution to optimize equipment allocation and trailer assignments.

This report was prepared using automated analysis of QuickManage API data.
Viscosity values are sourced from industrial databases and PubChem.
All data is based on the October 2025 – February 2026 operating period.